Linear Transformations

Linear Transformations

Solution 2

A linear transformation maps vectors by matrix multiplication:

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

Linearity means $T(\mathbf{u} + \mathbf{v}) = T\mathbf{u} + T\mathbf{v}$ and $T(k\mathbf{u}) = kT\mathbf{u}$.

For example, $M = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$. Find the image of $(1, 2)$.

Image

$$\begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \end{pmatrix}$$

The origin stays fixed because $M\mathbf{0} = \mathbf{0}$.

True or false?

Solution 3

Which statements are true?

1. A non-zero translation in the plane can be represented by a 2×2 matrix.
2. A linear map can change the dimension of the space.
3. Every linear transformation fixes the origin.

Decisions

1. False: a translation moves the origin.
2. True: for example,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

3. True: $M\mathbf{0} = \mathbf{0}$.

Examples

Solution 4

a. Find the matrix of $(x, y) \mapsto (x + 2y, 3x)$.

b. Find the matrix of $(x, y) \mapsto (-2y, 3x + y)$.

a

$$\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$$

b

$$\begin{pmatrix} 0 & -2 \\ 3 & 1 \end{pmatrix}$$

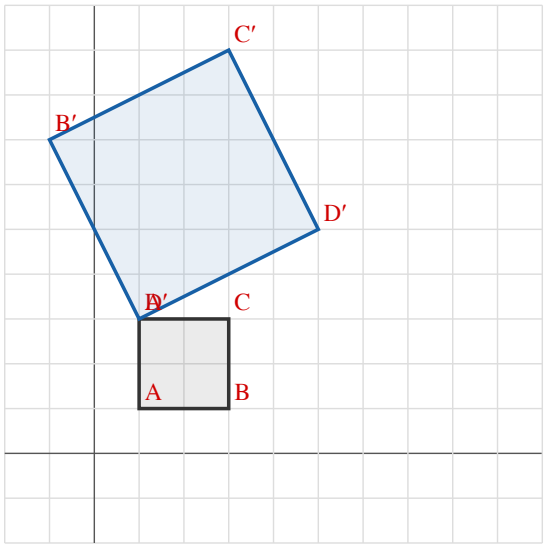
Transform the square $A(1,1), B(3,1), C(3,3), D(1,3)$ using $M = \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}$.

Vertices

$A'(1,3), B'(-1,7), C'(3,9), D'(5,5)$.

This reverses orientation and enlarges lengths by $\sqrt{5}$.

Sketch



Original: black · Image: blue · Equal axis scales

Exercise 7A Pure Core 1 - P129-130 — Solution 5

1. Which maps are linear?

a. $(x, y) \mapsto (2x, y + 1)$

b. $(x, y) \mapsto (x^2, y)$

c. $(x, y) \mapsto (2x + y, x + xy)$

d. $(x, y) \mapsto (y, -x)$

e. $(x, y) \mapsto (y + 3, x + 3)$

f. $(x, y) \mapsto (2x, 3y - 2x)$

a

a. No: constant term.

b. No: squared term.

c. No: product xy.

d. Yes: matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

e. No: constant terms.

f. Yes: matrix $\begin{pmatrix} 2 & 0 \\ -2 & 3 \end{pmatrix}$.

Exercise 7A Pure Core 1 - P129-130 — Solution 6

4a. Find the matrix of $(x, y) \mapsto (y + 2x, -y)$.

4b. Find the matrix of $(x, y) \mapsto (-y, x + 2y)$.

a

$\begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix}$

b

$\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$

Exercise 7A Pure Core 1 - P129-130 — Solution 7

7. A rectangle has vertices $A(2,1), B(4,1), C(4,2), D(2,2)$. Apply $M = -I$, sketch its image, and describe the transformation.

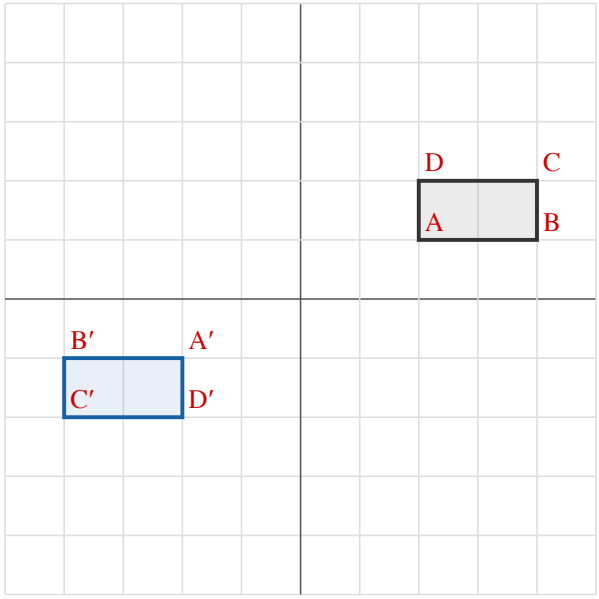
7

$A'(-2,-1),\ B'(-4,-1)$

$C'(-4,-2),\ D'(-2,-2).$

Rotation through $180°$ about the origin.

Sketch



Original: black · Image: blue · Equal axis scales

Exercise 7A Pure Core 1 - P129-130 — Solution 8

9. Transform the square with vertices $(-1,0), (-3,0), (-3,2), (-1,2)$ using $M = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$. Sketch and describe the image.

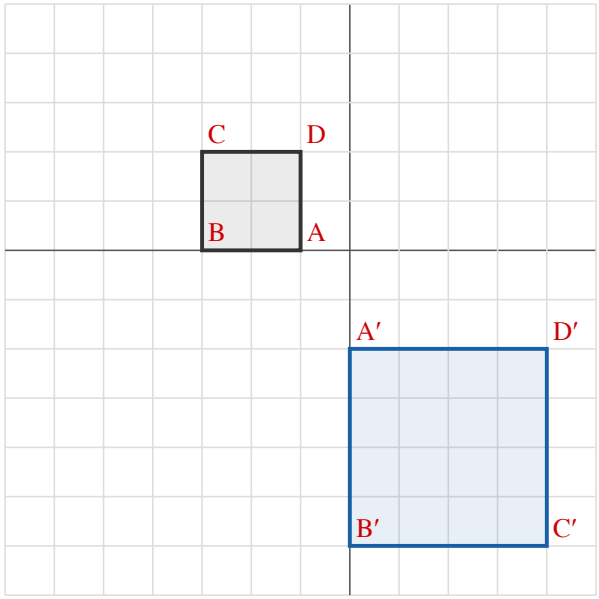
9

The images are

$(0,-2), (0,-6), (4,-6), (4,-2).$

Reflection in $y = x$, combined with enlargement of scale factor 2 about the origin.

Sketch



Original: black · Image: blue · Equal axis scales

Determining a matrix for a transformation

Solution 9

The columns of a transformation matrix are the images of the basis vectors.

$$\begin{pmatrix}1\\0\end{pmatrix} \mapsto \begin{pmatrix}a\\c\end{pmatrix}, \quad \begin{pmatrix}0\\1\end{pmatrix} \mapsto \begin{pmatrix}b\\d\end{pmatrix}$$

Therefore the matrix is $\begin{pmatrix}a & b\\c & d\end{pmatrix}$.

Find the matrix of reflection in the y-axis.

Reflection

$$\begin{pmatrix}1\\0\end{pmatrix} \mapsto \begin{pmatrix}-1\\0\end{pmatrix}$$

$$\begin{pmatrix}0\\1\end{pmatrix} \mapsto \begin{pmatrix}0\\1\end{pmatrix}$$

$$M = \begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}$$

Further Examples

Solution 10

A rotation through θ anticlockwise about the origin sends the basis vectors to

$$\begin{pmatrix}\cos\theta\\\sin\theta\end{pmatrix}$$

and

$$\begin{pmatrix}-\sin\theta\\\cos\theta\end{pmatrix}$$

Rotation matrix

$$R_\theta = \begin{pmatrix}\cos\theta & -\sin\theta\\\sin\theta & \cos\theta\end{pmatrix}$$

90 degrees

$$R_{90^\circ} = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}$$

Interactive transformation exploration

Solution 11

Explore how the images of the two basis vectors determine a transformation. Predict the effect before changing the matrix.

Open the interactive GeoGebra activity

Record your observations and test them using a matrix calculation.

Test Your Understanding So Far

Solution 12

1. Find the matrix of reflection in $y = x$.

1

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

2. Find the matrix of rotation 270° anticlockwise about the origin.

2

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Further Test Your Understanding

Solution 13

Edexcel FP1 January 2009, Q10. Describe the single transformation represented by

$$C = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Transformation

Rotation through $45°$ anticlockwise about the origin.

Invariant points and lines

Solution 14

Invariant points

An invariant point maps to itself. Solve $M\mathbf{x} = \mathbf{x}$.

For reflection in $y = x$, every point on $y = x$ is invariant.

For a non-identity rotation, only the origin is invariant.

Invariant lines

An invariant line maps onto itself; its individual points need not stay fixed.

Reflection in $y = x$ preserves $y = x$ and every line $y = -x + c$.

A $180°$ rotation preserves every line through the origin. Other non-identity rotations have no invariant lines.

Line of invariant points

Solution 15

Find the invariant points of $M = \begin{pmatrix} 5 & 4 \\ 2 & 3 \end{pmatrix}$.

Equations

$5x + 4y = x, \quad 2x + 3y = y$

$x + y = 0$

Result

Every point on $y = -x$ is fixed. The line is pointwise invariant.

Check by substituting $(x, -x)$ into the transformation.

Finding invariant lines – worked example in class

Solution 16

Find the invariant lines of $M = \begin{pmatrix} 3 & 2 \\ 5 & 6 \end{pmatrix}$.

Start with $y = mx + c$.

Images

$x' = (3 + 2m)x + 2c$

$y' = (5 + 6m)x + 6c$

Invariant lines

For $y' = mx' + c$, compare coefficients:

$5 + 6m = m(3 + 2m)$

$6c = (2m + 1)c$

$(2m - 5)(m + 1) = 0$

Hence $y = \frac{5}{2}x + c$ for any c, or $y = -x$.
No vertical line is invariant.

6. Let $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

a. Describe both transformations.

b. Find the image of $(3, 2)$ under A.

c. If $B(a, b) = (a - 3b, 2a - 2b)$, find a, b.

a

a. Reflection in the x-axis; rotation $270°$ anticlockwise about the origin.

b. $(3, -2)$.

c. $b = a - 3b$, $-a = 2a - 2b$, so $a = b = 0$.

7. Let $M = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$.

a. Describe M.

b. If $M(p, q) = (-\sqrt{2}, -2\sqrt{2})$, find p, q.

c. Find M^3.

d. Describe M^3.

a

a. Rotation $225°$ anticlockwise.

b. $-p + q = -2$, $-p - q = -4$: $p = 3, q = 1$.

c. $M^3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

d. Rotation $315°$ anticlockwise. Both centres are the origin.

8. Describe $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and find A^{50}.

8

Reflection in $y = x$. Since $A^2 = I$, $A^{50} = I$.

9. A rotation has matrix $\begin{pmatrix} a & b \\ c & -1/2 \end{pmatrix}$. Find the possible matrices.

9

$a = -\frac{1}{2}$, and the angle is $120°$ or $240°$.

$\frac{1}{2} \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}$

or

$\frac{1}{2} \begin{pmatrix} -1 & \sqrt{3} \\ -\sqrt{3} & -1 \end{pmatrix}$

10. A rotation of $270°$ clockwise about the origin maps (a, b) to $(a - 5b, 4b)$. Find the possible points.

10

$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

$-b = a - 5b, \quad a = 4b$

Every point $(4b, b)$, where b is real, satisfies the conditions.

Challenge

Derive the general rotation matrix using the images of $(1, 0)$ and $(0, 1)$.

Challenge

$(1, 0) \mapsto (\cos\theta, \sin\theta)$

$(0, 1) \mapsto (-\sin\theta, \cos\theta)$

$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

Describe these transformations:

$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

$B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

Examples

A: enlargement, scale factor 3, centre the origin.

B: stretch parallel to the x-axis, factor 2.

For positive a, b, $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ stretches by factor a parallel to the x-axis and factor b parallel to the y-axis.

A negative factor also reverses that coordinate; a zero factor collapses a dimension.

Area under a linear transformation

Solution 21

The triangle with vertices $(1,1),(1,2),(2,2)$ is transformed by $M=\begin{pmatrix}4&0\\0&3\end{pmatrix}$. Find its image and compare the areas.

Areas

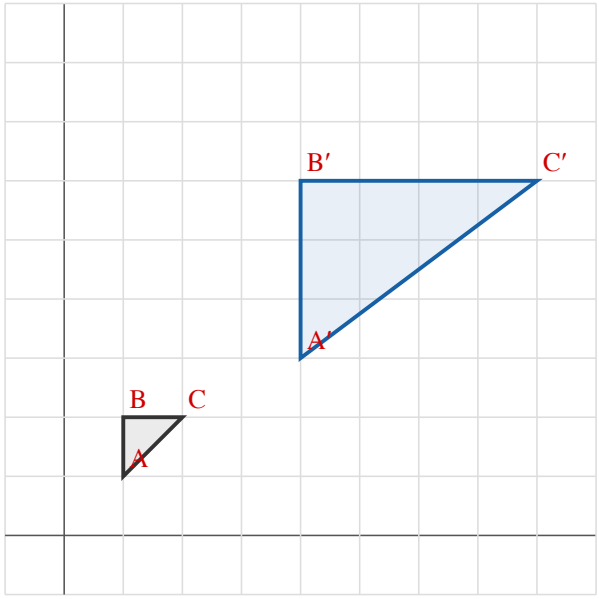
Images: $(4,3),(4,6),(8,6)$.

Original area $=\frac{1}{2}$

Image area $=\frac{1}{2}(4)(3)=6$

The area scale factor is $12=\det M$.

Sketch



Original: black · Image: blue · Equal axis scales

Area scale factor

Solution 22

Area scale factor

Image area $=|\det M|\times$ original area

A negative determinant reverses orientation. Find the image areas below.

a. Area 4, $M=\begin{pmatrix}1&2\\3&4\end{pmatrix}$.

b. Area 3, $M=\begin{pmatrix}2&0\\9&4\end{pmatrix}$.

a

$4|4-6|=8$

b

$3|8|=24$

c. Area 9, $M=\begin{pmatrix}5&3\\-2&-1\end{pmatrix}$.

d. Area 1, $M=\begin{pmatrix}-5&2\\-4&-2\end{pmatrix}$.

c

$9|-5+6|=9$

d

$1|10+8|=18$

Explore area scale factors

Solution 23

Explore how a matrix changes area. Compare the area scale factor with the absolute value of its determinant.

Open the interactive GeoGebra activity

Record your observations and test them using a matrix calculation.

Test Your Understanding

Solution 24

Edexcel January 2011, Q8. A shape is transformed by $A = \begin{pmatrix} 2 & -2 \\ -1 & 3 \end{pmatrix}$. Its image has area 72. Find its original area.

Area

$\det A = 6 - 2 = 4$

Original area $= 72/4 = 18$

5. Describe $M = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$. A triangle of area k has image area 24. Find k.

5

Stretches of factors 2 and 3 parallel to the coordinate axes, together with reflection in the x-axis.

$|\det M| = 6, \quad k = 24/6 = 4$

8. The triangle $(a, 1), (4, 1), (4, 3)$ is transformed by $2\sqrt{5}I$. Describe the transformation and find a if its image area is 60.

8

Enlargement, factor $2\sqrt{5}$, centre the origin.

$20|4 - a| = 60$

$a = 1$ or 7

9. Let $M = \begin{pmatrix} p & 1 \\ p & q \end{pmatrix}$, where $q > 0$, and $M^2 = 6I$. Find p, q.

9

$M^2 = \begin{pmatrix} p^2 + p & p + q \\ p^2 + pq & p + q^2 \end{pmatrix}$

$p + q = 0$, and $p^2 + p = 6$, giving $p = 2$ or -3. Since $q > 0$, $p = -3, q = 3$.

11. A pentagon of area 12 is transformed by $M = \begin{pmatrix} -1 & \sqrt{2} \\ -\sqrt{2} & -1 \end{pmatrix}$. Find the image area.

11

$\det M = 1 + 2 = 3$

Image area $= 36$

12. The triangle with vertices $(k, 1), (4, 1), (4, k)$, where k is an integer, is transformed by $M = \begin{pmatrix} 4 & -1 \\ k & 2 \end{pmatrix}$. Its image area is 10. Find k.

Area equation

$\frac{1}{2}|(4-k)(k-1)(k+8)| = 10$

Solution

If $(k-4)(k-1)(k+8) = -20$,

$(k-2)(k^2+5k-26) = 0$

only $k = 2$ is an integer. For the positive sign, the cubic is

$k^3 + 3k^2 - 36k + 12 = 0$

Any integer root must divide 12; none of $\pm1, \pm2, \pm3, \pm4, \pm6, \pm12$ works. Hence $k = 2$.

13. Rotate the triangle $(0, 0), (7, 7), (3, -2)$ through $45°$ anticlockwise about the origin. Find its image, determinant and area.

Images

$R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Images: $(0, 0), (0, 7\sqrt{2}), (5/\sqrt{2}, 1/\sqrt{2})$.

Area

$\det R = 1$

Area $= \frac{1}{2}|7(-2) - 7(3)| = \frac{35}{2}$

The image has the same area.

Combining transformations

Apply A first, then B:

$\mathbf{x} \mapsto A\mathbf{x} \mapsto B(A\mathbf{x})$

Order

Combined matrix $= BA$

Matrix multiplication is associative: $B(A\mathbf{x}) = (BA)\mathbf{x}$.

It is generally not commutative: $BA \neq AB$. The rightmost transformation acts first.

1. Reflect in $y = x$, then stretch parallel to the x-axis by factor 4. Find the combined matrix.

2. Rotate 90° anticlockwise, then reflect in $y = x$. Describe the combined transformation.

1

$$\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$$

2

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Reflection in the x-axis.

Test Your Understanding

Solution 31

Edexcel January 2013, Q4. Let P represent rotation 90° anticlockwise about the origin and Q reflection in $y = -x$. Find the matrix and transformation for P followed by Q.

Combined transformation

$$P = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

$$QP = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Reflection in the y-axis.

Exercise 7D Pure Core 1 - P141-143

Solution 32

5. Show that reflection in the y-axis followed by reflection in $y = -x$ is a rotation. Give its angle and centre.

5

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Rotation 90° anticlockwise about the origin.

7. Let $P = \begin{pmatrix} -4 & 0 \\ 0 & 2 \end{pmatrix}$ and $Q = kI$.
Find PQ, describe the transformation, and compare PQ with QP.

7

$$PQ = QP = \begin{pmatrix} -4k & 0 \\ 0 & 2k \end{pmatrix}$$

The coordinates are multiplied by $-4k$ and $2k$. For $k > 0$, these are stretches of factors $4k, 2k$ with reflection in the y-axis. For $k < 0$, reflection is in the x-axis, with factors $4|k|, 2|k|$. For $k = 0$, all points map to the origin.

8. Find and describe A^2, where $A = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$.
Generalise for $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$.

a

$$A^2 = \begin{pmatrix} 9 & 0 \\ 0 & 16 \end{pmatrix}$$

Stretches by factors 9 and 16 parallel to the x- and y-axes.

b

$$B^2 = \begin{pmatrix} a^2 & 0 \\ 0 & b^2 \end{pmatrix}$$

Scale the coordinates by a^2, b^2. A zero factor collapses that coordinate.

10. An enlargement of negative scale factor k, followed by a rotation of angle θ, has matrix

$$M = \frac{3}{\sqrt{2}}\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$$

Find k and $0 \leq \theta < 360°$.

10

$k = -3, \quad \theta = 45°$

11. Let $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $B = 5I$. A shape is transformed by B then A; its image area is 75. Find the original area.

11

$|\det(AB)| = 25$

Original area $= 75/25 = 3$

13. Show that applying $A = \begin{pmatrix} k & \sqrt{3} \\ \sqrt{3} & -k \end{pmatrix}$ twice is an enlargement.

13

$A^2 = (k^2 + 3)I$

Enlargement, scale factor $k^2 + 3$, centre the origin.

14. Generalise for $P = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$.

14

$P^2 = (a^2 + b^2)I$

Enlargement, scale factor $a^2 + b^2$, centre the origin. If $a = b = 0$, the plane collapses to the origin.

Transformations in three dimensions

The three columns are the images of $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Find the matrix of reflection in the plane $z = 0$.

Reflection

$(x, y, z) \mapsto (x, y, -z)$

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

The x- and y-coordinates stay fixed; the z-coordinate changes sign.

Rotations in three dimensions

Use the right-hand convention: positive rotation is anticlockwise when looking towards the origin from the positive end of the axis.

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Explore three-dimensional transformations

Solution 39

Explore rotations and reflections in three dimensions. Follow the images of the basis vectors.

[Open the interactive GeoGebra activity]

Record your observations and test them using a matrix calculation.

Test Your Understanding

Solution 40

Describe the transformation and find the image of $(-1, -2, 1)$:

$$M = \begin{pmatrix} \sqrt{3}/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & \sqrt{3}/2 \end{pmatrix}$$

Transformation

Rotation $30°$ about the y-axis, in the positive convention.

Image

$$\begin{pmatrix} (1-\sqrt{3})/2 \\ -2 \\ (1+\sqrt{3})/2 \end{pmatrix}$$

Exercise 7E Pure Core 1 - P146-147

Solution 41

3. Let $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$.

a. Describe M.

b. Find the image of $A(3, -1, 4)$.

c. If $(a, -a, 2a - 1)$ maps to $(a, a - 5, -a)$, find a.

a

a. Rotation 90° about the x-axis in the positive convention.

b

$A' = (3, -4, -1)$

c

$1 - 2a = a - 5 \Rightarrow a = 2$

Exercise 7E Pure Core 1 - P146-147

Solution 42

4. P rotates 120° anticlockwise about the z-axis (viewed from its positive end).

a. Find P.

b. Find the image of $(3, -1, 0)$.

c. Find the image of $(k, 0, k)$.

a

$$P = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} & 0 \\ \sqrt{3} & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

b

$$\begin{pmatrix} (-3 + \sqrt{3})/2 \\ (3\sqrt{3} + 1)/2 \\ 0 \end{pmatrix}$$

c

$$\begin{pmatrix} -k/2 \\ k\sqrt{3}/2 \\ k \end{pmatrix}$$

Exercise 7E Pure Core 1 - P146-147 Solution 43

5. A reflects in $x = 0$; B reflects in $y = 0$.

a. Find A, B.

b. Find the image P' of $P(a, b, c)$ under A.

c. Transform P' using B.

a

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

b

$$P' = (-a, b, c)$$

c

$$P'' = (-a, -b, c)$$

Exercise 7E Pure Core 1 - P146-147 Solution 44

6. Describe M and find the image of $(k, -k, 0)$:

$$M = \frac{1}{2}\begin{pmatrix} -\sqrt{3} & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & -\sqrt{3} \end{pmatrix}$$

a

Rotation $210°$ about the y-axis in the positive convention.

b

$$\begin{pmatrix} -k\sqrt{3}/2 \\ -k \\ k/2 \end{pmatrix}$$

Inverse matrices for inverse transformations

Solution 45

Inverse transformations

If $\mathbf{y} = M\mathbf{x}$ and M is invertible, then $\mathbf{x} = M^{-1}\mathbf{y}$.

$M = \begin{pmatrix} 4 & -1 \\ 3 & 1 \end{pmatrix}$ maps a triangle to $A'(4, 3), B'(4, 10), C'(-4, -3)$. Find its original vertices.

Inverse

$M^{-1} = \frac{1}{7}\begin{pmatrix} 1 & 1 \\ -3 & 4 \end{pmatrix}$

Vertices

Apply M^{-1} to each image:

$A = (1, 0), \quad B = (2, 4), \quad C = (-1, 0)$

Test Your Understanding

Solution 46

Edexcel June 2012, Q9 (selected parts).

$M = \begin{pmatrix} 3 & 4 \\ 2 & -5 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

a. Find $\det M$.

e. Describe A.

f. A followed by B is equivalent to M. Find B.

a

$\det M = -15 - 8 = -23$

e

Rotation 90° anticlockwise about the origin.

f

$M = BA \Rightarrow B = MA^{-1}$

$B = \begin{pmatrix} -4 & 3 \\ 5 & 2 \end{pmatrix}$

Exercise 7F Pure Core 1 - P148-150 — Solution 47

4. $A = \begin{pmatrix} 1 & 2 \\ -3 & -2 \end{pmatrix}$ maps P to $(5, 8)$.

a. Find P.

b. Applying A then B gives reflection in $y = -x$. Find B.

a

$A^{-1} = \frac{1}{4}\begin{pmatrix} -2 & -2 \\ 3 & 1 \end{pmatrix}$

$P = (-13/2, 23/4)$

b

$BA = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

$B = \frac{1}{4}\begin{pmatrix} -3 & -1 \\ 2 & 2 \end{pmatrix}$

Exercise 7F Pure Core 1 - P148-150 — Solution 48

5. Let $E = 4I$.

a. Describe E.

b. Find E^{-1}.

c. A triangle has image vertices $(4, 6), (9, 7), (3, 1)$. Find the original vertices.

a

Enlargement, factor 4, centre the origin.

b

$E^{-1} = \frac{1}{4}I$

c

$(1, 3/2), \quad (9/4, 7/4), \quad (3/4, 1/4)$

6. Let $M = \text{diag}(a, b)$, with $a, b \neq 0$.
   a. Find M^{-1}.
   b. Find the point mapping to $(-6, 8)$.

6
$M^{-1} = \text{diag}(1/a, 1/b)$
$(-6/a, 8/b)$

7. Let $R = \frac{1}{2}\begin{pmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{pmatrix}$.
   a. Describe R.
   b. Find the point mapping to $((1 - 2\sqrt{3})/2, (2 + \sqrt{3})/2)$.

7
Rotation $30°$ clockwise about the origin.
$R^{-1} = \frac{1}{2}\begin{pmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{pmatrix}$
Applying the inverse gives $(-2, 1)$.

8. $A = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. A triangle is transformed by AB. Find the matrix that maps the image back.

8
$AB = \begin{pmatrix} -4 & -2 \\ -3 & -1 \end{pmatrix}$
$(AB)^{-1} = \begin{pmatrix} 1/2 & -1 \\ -3/2 & 2 \end{pmatrix}$

9. $A = \begin{pmatrix} 6 & -2 \\ -4 & 1 \end{pmatrix}$ maps P to (a, b). Find P.

9
$A^{-1} = \frac{1}{2}\begin{pmatrix} -1 & -2 \\ -4 & -6 \end{pmatrix}$
$P = (-a/2 - b, -2a - 3b)$

10. $M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & -2 \\ 1 & 3 & -1 \end{pmatrix}$ maps (a, b, c) to $(3, 2, -1)$. Find M^{-1} and a, b, c.

10

$$M^{-1} = \begin{pmatrix} 2 & 3/2 & -1 \\ -1 & -1 & 1 \\ -1 & -3/2 & 1 \end{pmatrix}$$

$$(a, b, c) = (10, -6, -7)$$

12. $A = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 1 & 2 \\ 2 & 0 & -1 \end{pmatrix}$. Applying A then B gives reflection in the xy-plane. Find B.

12

$$BA = \text{diag}(1, 1, -1)$$

$$B = \text{diag}(1, 1, -1)A^{-1}$$

$$B = \frac{1}{10}\begin{pmatrix} -1 & 3 & 7 \\ 3 & 1 & -1 \\ 2 & -6 & -4 \end{pmatrix}$$